

BEALE AFB, CA, USA

WMO: 724837

Lat: 39.133N

Lon: 121.433W

Elev: 34

StdP: 100.91

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 93216

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-0.5	1.0	-6.3	2.2	8.9	-4.3	2.6	8.5	13.4	11.7	11.6	12.1	1.6	90	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	38.4	21.2	36.9	20.7	35.2	20.0	22.8	35.4	21.9	34.0	21.1	33.0	3.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.3	13.3	28.2	17.4	12.5	26.7	16.4	11.7	26.0	67.3	35.4	63.7	34.3	60.7	33.0	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	8.7	7.6	DB	-3.4	41.0	2.0	0.8	-4.8	41.6	-5.9	42.1	-7.1	42.5	-8.5	43.1	(4)
(5)			WB	-3.7	24.5	2.0	1.2	-5.2	25.4	-6.3	26.1	-7.5	26.7	-8.9	27.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	8.9	10.8	12.9	15.5	19.2	23.3	25.9	25.2	23.2	17.8	12.3	8.3
	DBStd	6.83	2.69	2.69	2.83	3.36	3.60	3.67	2.59	2.60	3.02	2.80	3.09	2.98
	HDD10.0	163	52	21	7	2	0	0	0	0	0	0	12	69
	HDD18.3	1341	292	211	168	97	32	4	0	0	3	43	181	311
	CDD10.0	2709	18	43	98	167	284	400	493	471	396	242	82	17
	CDD18.3	846	0	0	1	12	57	154	234	212	148	27	1	0
	CDH23.3	11409	0	1	31	241	823	2105	3137	2762	1851	442	17	0
	CDH26.7	5968	0	0	3	64	339	1137	1811	1540	939	134	1	0
Wind	WSAvg	3.0	2.9	3.4	3.4	3.4	3.4	3.3	3.0	2.8	2.7	2.7	2.6	3.0
Precipitation	PrecAvg	682	134	114	106	46	34	11	1	2	5	34	59	138
	PrecMax	1118	360	310	340	158	120	41	4	24	18	103	141	311
	PrecMin	225	13	1	7	0	0	0	0	0	0	0	0	15
	PrecStd	204	91	81	94	39	33	13	1	5	6	28	41	89
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.2	22.1	26.8	31.2	36.0	40.2	40.2	40.1	38.1	33.0	25.3	17.9
		MCWB	11.6	14.3	16.5	17.7	20.8	22.1	21.8	21.1	21.1	17.6	15.2	12.1
	2%	DB	16.7	20.0	23.8	28.5	32.8	37.7	38.2	37.7	35.8	30.1	23.0	16.1
		MCWB	11.7	13.7	15.1	17.2	18.6	21.1	21.1	20.9	19.6	16.9	14.4	11.8
	5%	DB	15.1	17.8	21.9	26.3	30.6	35.3	36.7	35.8	34.0	27.8	20.7	14.8
		MCWB	11.7	12.6	14.2	16.2	18.1	19.9	20.8	20.2	19.1	15.8	13.5	11.4
	10%	DB	13.8	16.2	19.7	23.8	27.9	32.9	34.9	34.0	32.3	25.9	18.1	13.4
		MCWB	11.0	12.0	13.3	15.2	17.1	19.1	20.5	19.7	18.5	15.1	12.8	10.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	16.2	17.7	20.0	22.2	24.1	24.2	23.1	22.5	19.4	17.0	15.1
		MCDB	15.7	19.6	23.6	27.9	32.5	36.8	36.4	36.9	35.4	28.7	21.0	15.9
	2%	WB	13.6	15.0	16.1	18.0	20.5	22.5	22.9	21.9	20.9	18.0	15.8	13.8
		MCDB	15.1	17.9	21.8	26.3	30.5	34.3	35.6	35.1	32.6	26.8	20.1	14.8
	5%	WB	12.5	13.9	15.0	16.8	19.0	21.3	21.9	21.1	20.0	17.1	14.9	12.6
		MCDB	13.9	16.7	19.9	24.6	28.7	32.7	34.1	33.7	31.8	25.0	18.7	13.6
	10%	WB	11.6	12.8	14.0	15.8	17.7	20.1	21.0	20.3	19.1	16.2	13.8	11.5
		MCDB	13.0	15.4	18.4	22.5	26.4	31.2	33.2	32.5	30.5	23.3	17.3	12.6
Mean Daily Temperature Range	5% DB	MDBR	8.7	9.9	11.5	13.5	14.4	16.3	17.5	17.4	16.4	14.8	11.2	8.9
		MCDBR	11.4	13.2	15.8	18.0	18.2	19.2	19.0	19.0	19.2	18.2	15.1	10.7
		MCWBR	7.3	7.8	8.5	8.4	7.2	7.0	7.5	7.7	7.6	7.9	7.6	6.7
	5% WB	MCDBR	7.5	10.7	13.4	16.1	16.8	18.0	17.8	18.0	17.5	15.1	11.4	6.6
		MCWBR	5.5	6.8	7.9	8.3	7.4	6.8	7.6	7.6	7.3	7.0	6.4	5.0
Clear-Sky Solar Irradiance	taub	0.323	0.326	0.350	0.358	0.371	0.363	0.361	0.345	0.353	0.345	0.324	0.335	
	taud	2.505	2.505	2.429	2.401	2.391	2.421	2.415	2.454	2.440	2.462	2.510	2.454	
	Ebn at Noon	851	900	908	918	908	911	910	920	892	863	842	802	
	Edh at Noon	78	88	105	115	119	115	115	108	103	91	76	76	
All-Sky Solar Radiation	RadAvg	2.14	3.11	4.44	5.94	7.30	8.23	8.25	7.36	5.93	4.12	2.59	1.88	
	RadStd	0.39	0.42	0.52	0.56	0.48	0.36	0.26	0.25	0.22	0.30	0.21	0.33	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (41 neighbors)	+0.41	N/A	-1.13	N/A	N/A	N/A	N/A	-107	+137	+44

Nomenclature: See separate page